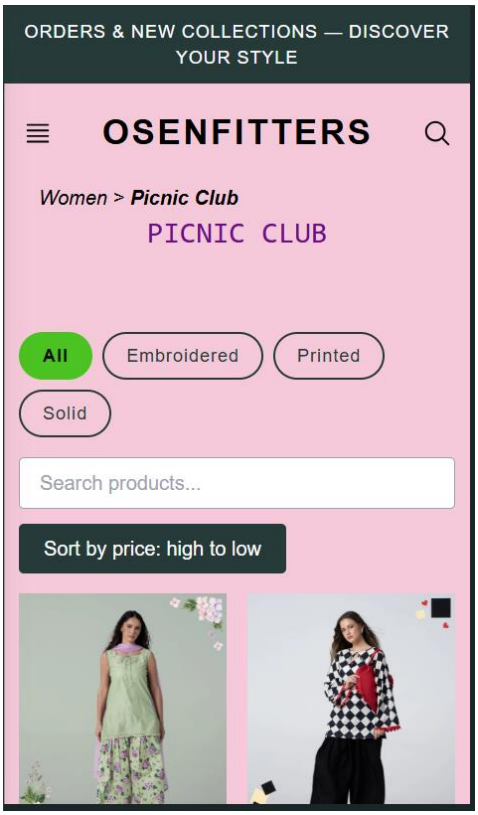
Web Technologies - Lab Mid Term

Name: Zainab Hussain
Registration No: FA24-BCS-115
N = 15, BP = 615px, H = 105, K = 3

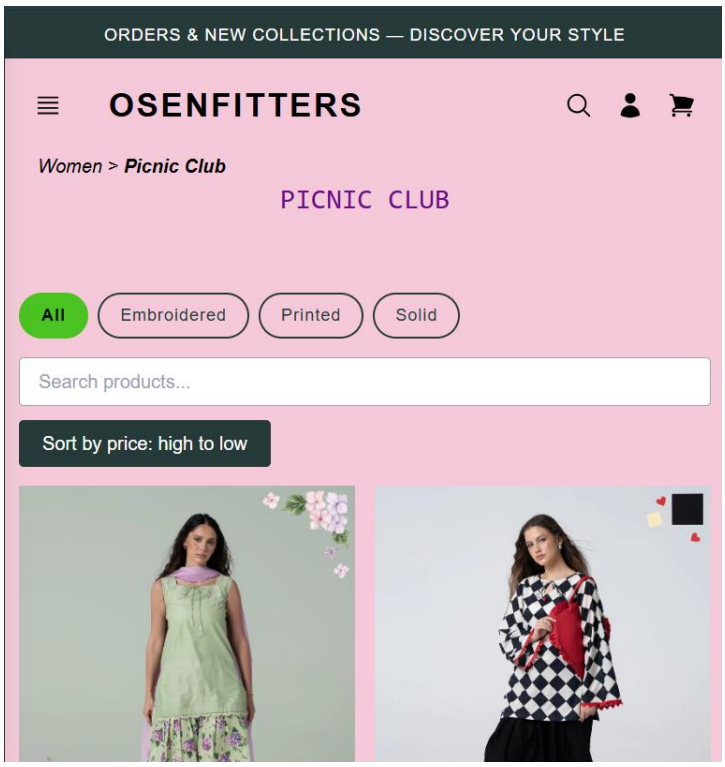
Task 1 - CSS-only category filter

Page: picnicclub.html (product listing page). Categories: Embroidered, Printed, Solid. The filter uses radio buttons and the :checked and ~ selectors, with no JavaScript.

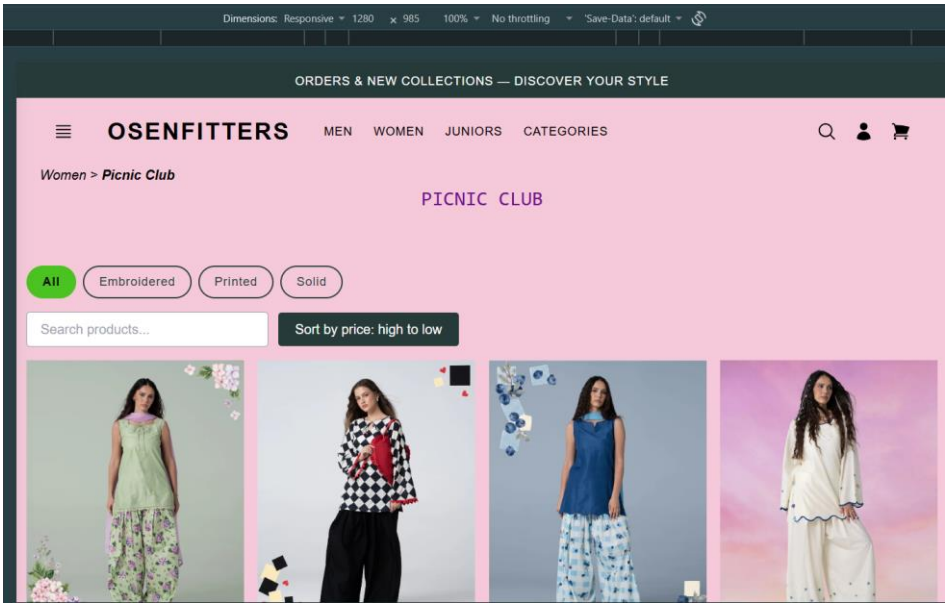
Screenshots



[screenshot Task 1 at 375px]



[screenshot of Task 1 at 615px (BP)]



[screenshot of Task 1 at 1280px]

Code added or changed

CSS (in the head of picnicclub.html):

```
/* Midterm: N = 15, H = (15 x 7) mod 360 = 105 */
```

```
:root {
  --accent: hsl(105, 70%, 45%);
}

/* ---------- Task 1: CSS-only category filter ---------- */
#shop {
  position: relative;
}
.filter-radio {
  position: absolute;
  width: 1px;
  height: 1px;
  opacity: 0;
  overflow: hidden;
}
.filter-label {
  display: inline-block;
  padding: 0.45rem 1.1rem;
  border: 2px solid #263b39;
  border-radius: 9999px;
  font-size: 0.875rem;
  letter-spacing: 1px;
  color: #263b39;
  cursor: pointer;
}
.filter-label:hover {
  border-color: var(--accent);
}

/* selected radio -> its label gets the accent colour */
#f-all:checked ~ .filter-bar label[for="f-all"],
#f-embroidered:checked ~ .filter-bar label[for="f-embroidered"],
#f-printed:checked ~ .filter-bar label[for="f-printed"],
#f-solid:checked ~ .filter-bar label[for="f-solid"] {
  background: var(--accent);
  border-color: var(--accent);
  color: #111111;
  font-weight: 700;
}

/* keyboard focus */
#f-all:focus-visible ~ .filter-bar label[for="f-all"],
#f-embroidered:focus-visible ~ .filter-bar label[for="f-embroidered"],
#f-printed:focus-visible ~ .filter-bar label[for="f-printed"],
#f-solid:focus-visible ~ .filter-bar label[for="f-solid"] {
  outline: 3px solid #111111;
  outline-offset: 2px;
}

/* hide cards that do not match the checked category (display:none -> no gaps) */
#f-embroidered:checked ~ #product-grid article:not([data-category="embroidered"]),
#f-printed:checked ~ #product-grid article:not([data-category="printed"]),
#f-solid:checked ~ #product-grid article:not([data-category="solid"]) {
  display: none;
}
```

HTML - radio buttons and labels (placed before the grid):

```
<div id="shop" class="mx-3">
      <!-- Task 1: radios must come BEFORE the grid (sibling selectors) -->
      <input type="radio" name="category" id="f-all" class="filter-radio" checked />
      <input type="radio" name="category" id="f-embroidered" class="filter-radio" />
      <input type="radio" name="category" id="f-printed" class="filter-radio" />
      <input type="radio" name="category" id="f-solid" class="filter-radio" />

      <div class="filter-bar flex flex-wrap gap-2 mb-4" role="group" aria-label="Filter by
category">
        <label for="f-all" class="filter-label">All</label>
        <label for="f-embroidered" class="filter-label">Embroidered</label>
        <label for="f-printed" class="filter-label">Printed</label>
        <label for="f-solid" class="filter-label">Solid</label>
      </div>
```

HTML - each product card got a data-category attribute (example):

```
<article data-category="printed">
        <img
          src="picnicclub/mochi.webp"
          aria-label="mochi article"
          class="hover:scale-110"
        />
        <p class="product-name">Mochi</p>
        <p class="text-xs font-semibold text-[#544f56]">PICNIC CLUB</p>
        <p class="product-price font-bold">PKR 10,350</p>
        <div class="w-4 h-4 bg-[#8ef690] mb-1 border-black border-b-2"></div>
        <button
          type="button"
          class="w-full bg-[#263b39] text-white py-2 px-3 mt-2 hover:bg-[#405b57] rounded"
        >
          Add to Cart
        </button>
      </article>
```

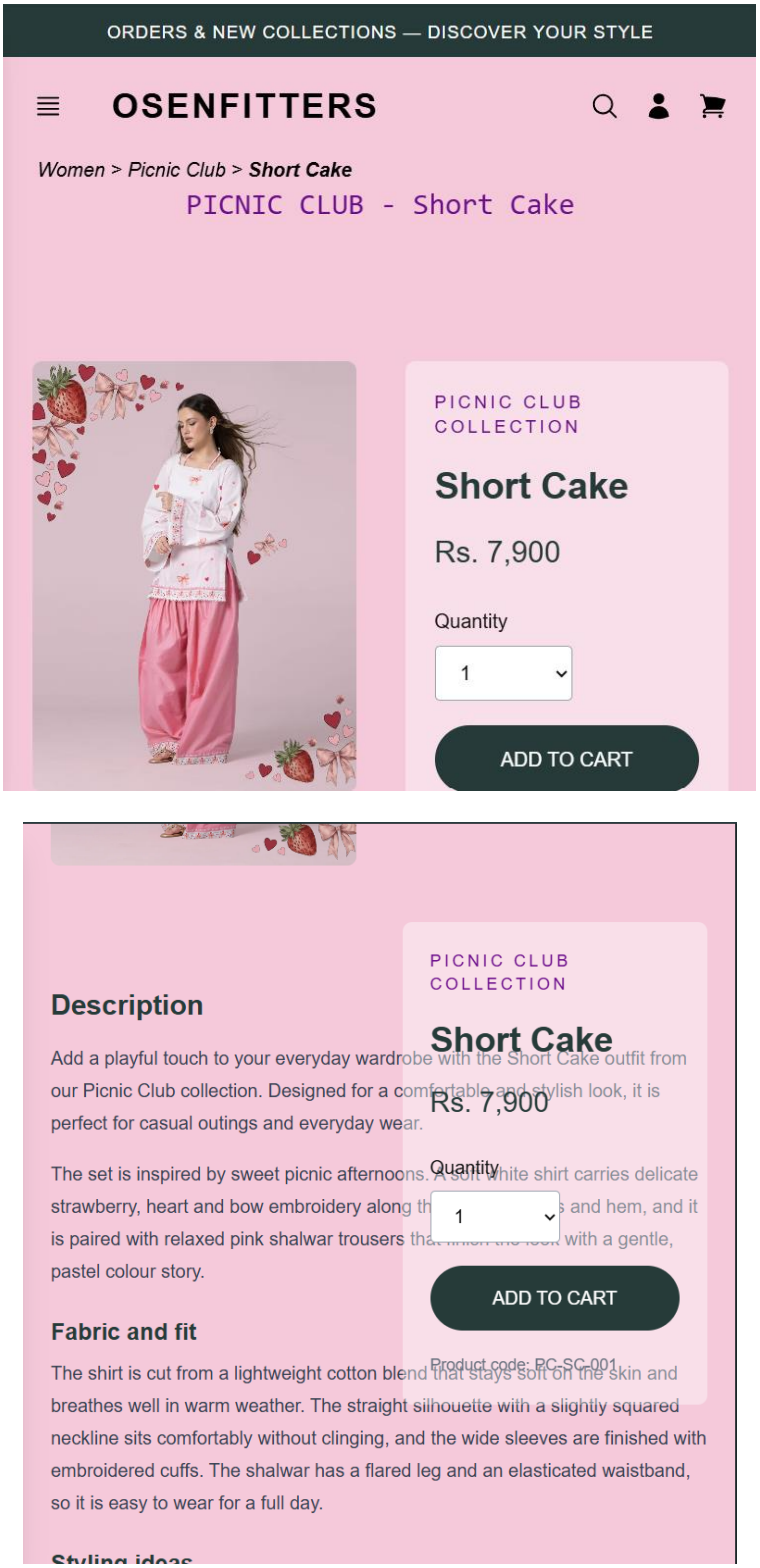
Task 2 - Product page layout with sticky details box

Page: shortcake.html (product detail page). CSS Grid with named areas; breakpoint 615px; sticky top = (15 mod 40) + 70 = 85px.

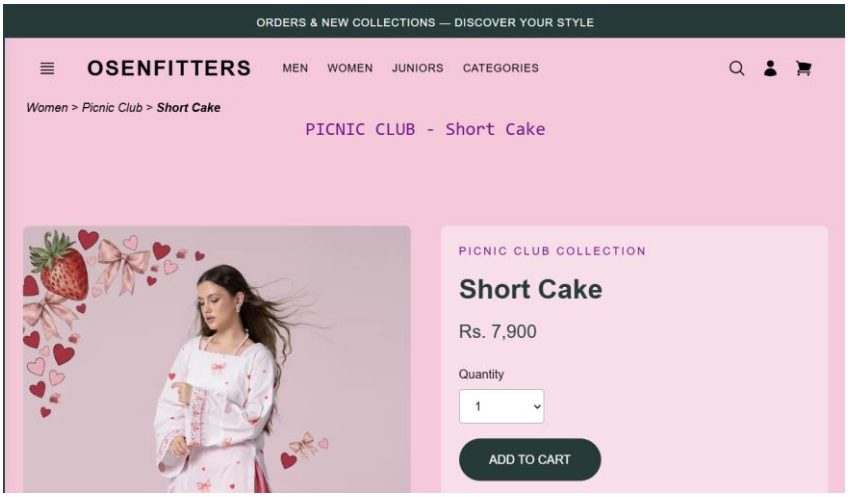
Screenshots

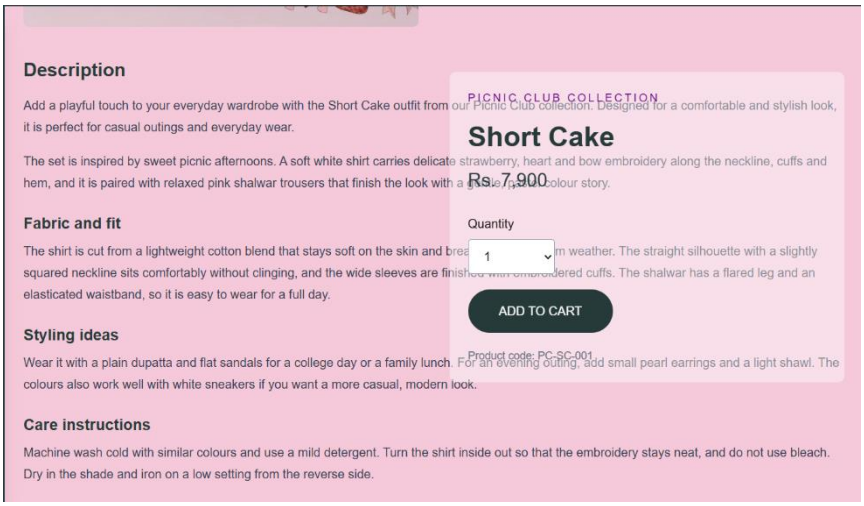


[screenshot of Task 2 at 375px]



[screenshot of Task 2 at 615px (BP)]





[screenshot of Task 2 at 1280px]

Code added or changed

CSS (in the head of shortcake.html):

```
/* Midterm: N = 15 -> BP = 600 + 15 = 615px
         sticky offset = (15 mod 40) + 70 = 85px */

      /* below BP: one column -> image, details, description */
      .product-layout {
        display: grid;
        grid-template-columns: 1fr;
        grid-template-areas:
          "image"
          "details"
          "description";
        gap: 2.5rem;
      }
      .product-image {
        grid-area: image;
      }
      .product-details {
```

```
      grid-area: details;
    }
    .product-description {
      grid-area: description;
    }

    /* from BP: image + details side by side, description full width */
    @media (min-width: 615px) {
      .product-layout {
        grid-template-columns: 1fr 1fr;
        grid-template-areas:
          "image details"
          "description description";
        align-items: start;
      }
      .product-details {
        position: sticky;
        top: 85px;
        align-self: start;
      }
    }
```

HTML structure:

```
<article
      id="short-cake-details"
      class="product-layout max-w-6xl mx-auto px-6 py-12"
    >
      <div class="product-image w-full">
        <img
          src="collectionpics/picnicclub.webp"
          alt="Short Cake from Picnic Club collection"
          class="w-full aspect-[3/4] object-cover rounded-lg"
        />
      </div>

      <div class="product-details flex flex-col items-start gap-5 bg-white/40 rounded-lg p-6">
        <p class="text-sm tracking-[3px] text-[#6c0d8c] uppercase">
          Picnic Club Collection
        </p>

        <h1 class="text-3xl md:text-4xl font-semibold text-[#263b39]">
          Short Cake
        </h1>

        <p class="text-2xl font-medium text-[#263b39]">Rs. 7,900</p>

        <!-- Quantity Selector -->
        <div class="mt-2">
          <label for="quantity" class="block mb-2 font-medium">
            Quantity
          </label>

          <select
            id="quantity"
            name="quantity"
            class="border border-gray-400 rounded px-4 py-3 w-28 bg-white"
          >
            <option value="1">1</option>
            <option value="2">2</option>
            <option value="3">3</option>
            <option value="4">4</option>
            <option value="5">5</option>
            <option value="6">6</option>
            <option value="7">7</option>
            <option value="8">8</option>
            <option value="9">9</option>
            <option value="10">10</option>
          </select>
        </div>

        <!-- Add to Cart Button -->
        <button
```

```
          type="button"
          class="w-full md:w-auto bg-[#263b39] text-white py-4 px-10 rounded-full hover:bg-
[#6c0d8c] transition duration-300 cursor-pointer"
        >
          ADD TO CART
        </button>

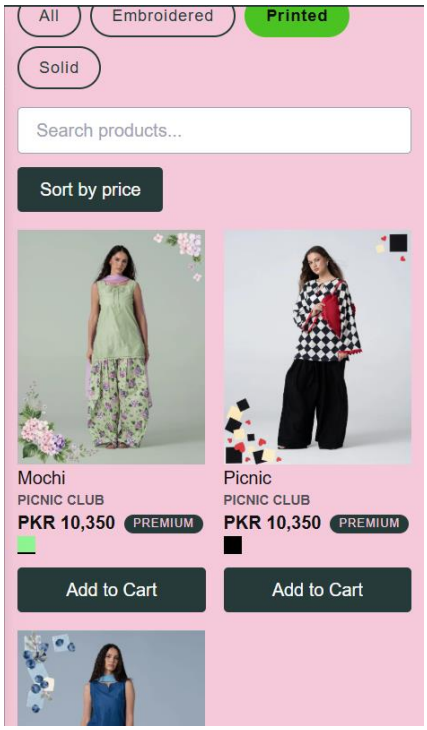
        <p class="text-sm text-gray-500">Product code: PC-SC-001</p>
      </div>
      ... (details box content unchanged) ...

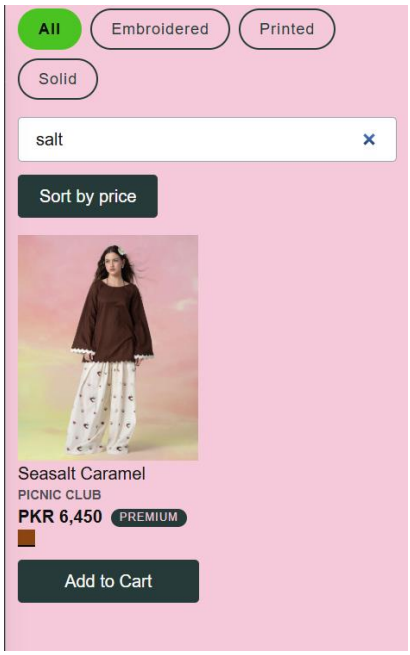
      <section class="product-description ...">
        <h2>Description</h2>
        ... (long description paragraphs) ...
      </section>
    </article>
```

Task 3 - JavaScript: live search and price sort

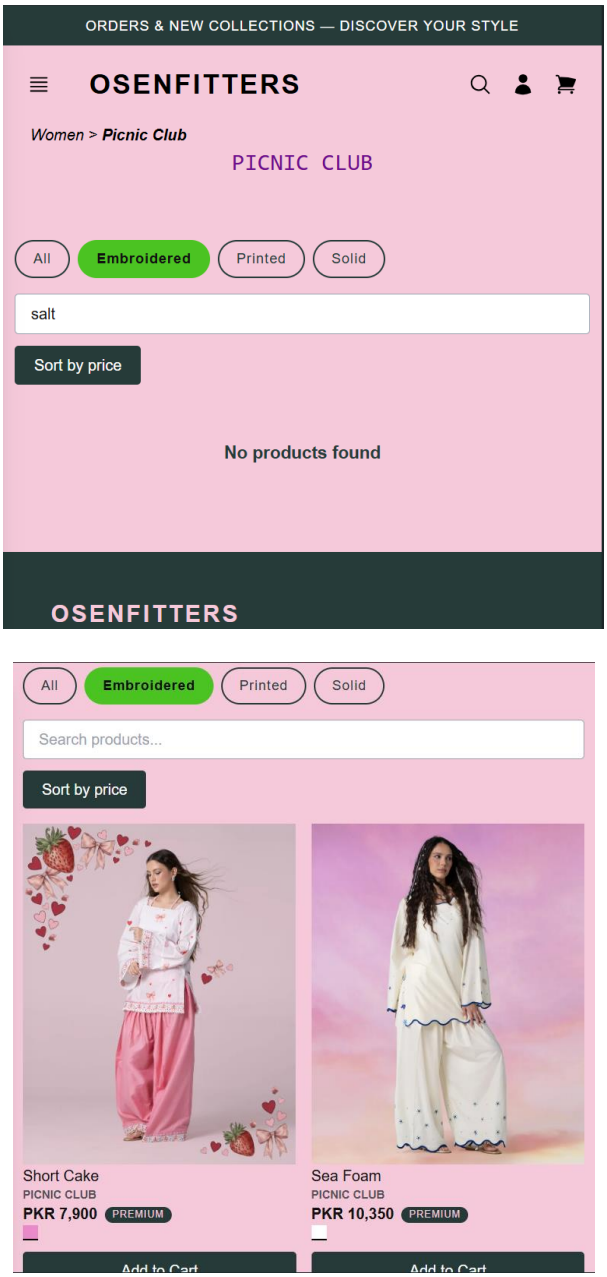
Files: script.js (new) linked at the bottom of picnicclub.html. Premium limit = 15 x 100 = 1500.

Screenshots

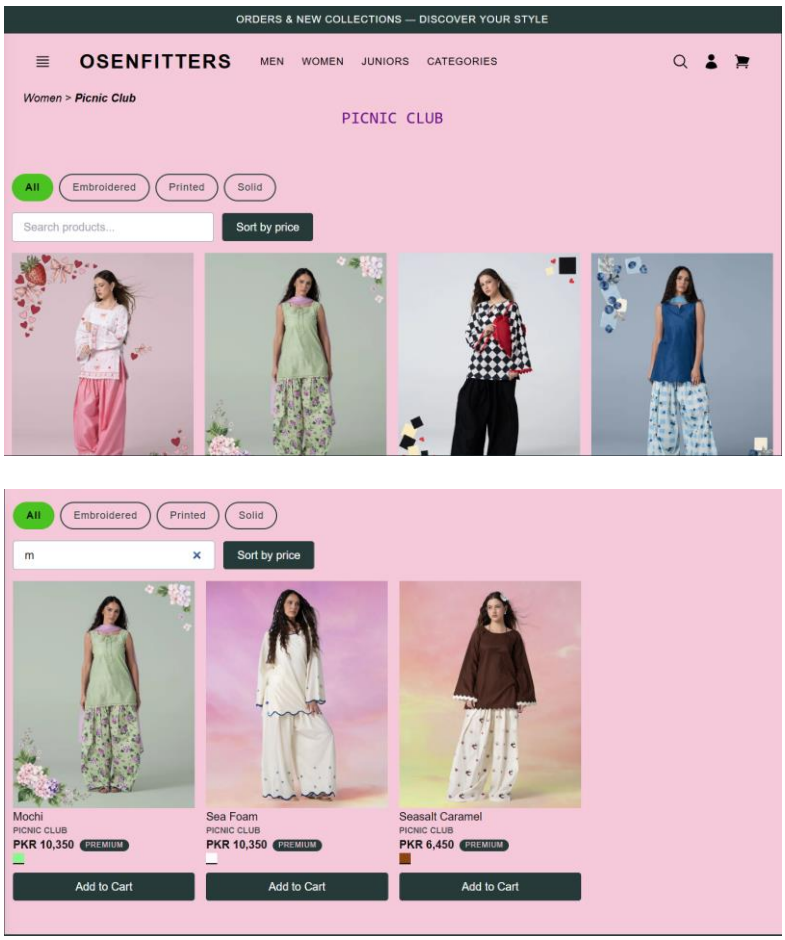




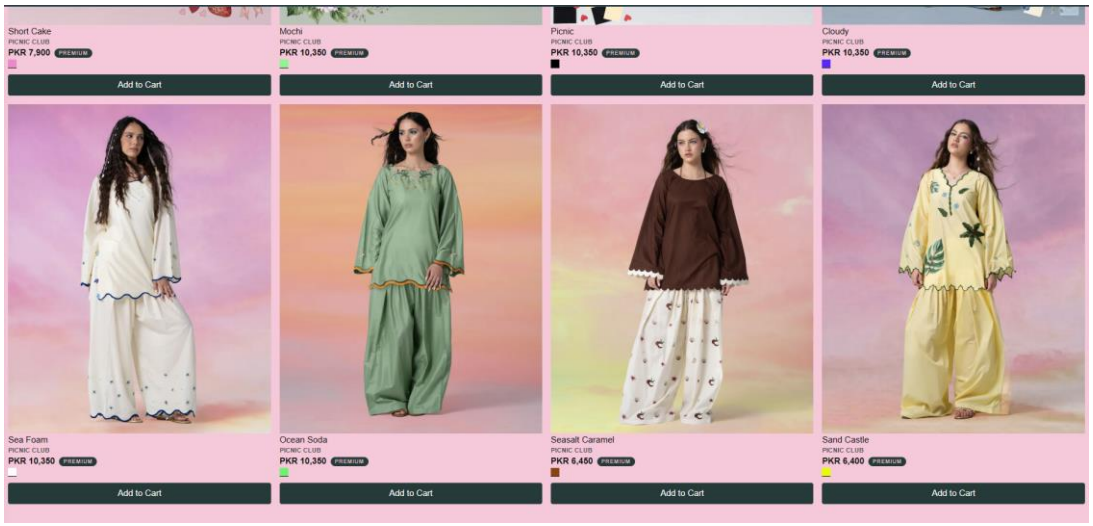
[screenshot of Task 3 at 375px]



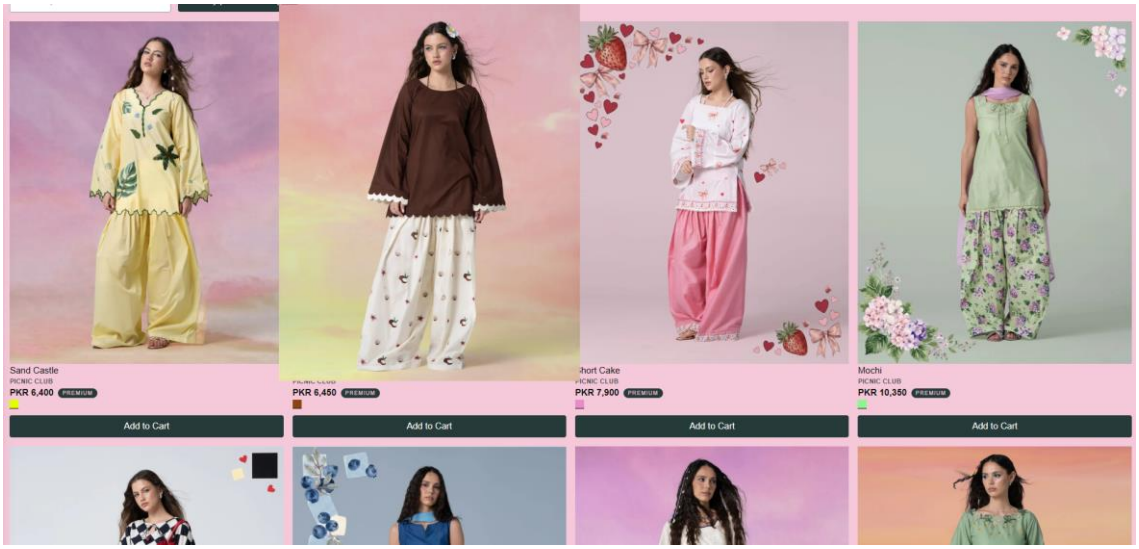
[screenshot of Task 3 at 615px (BP)]



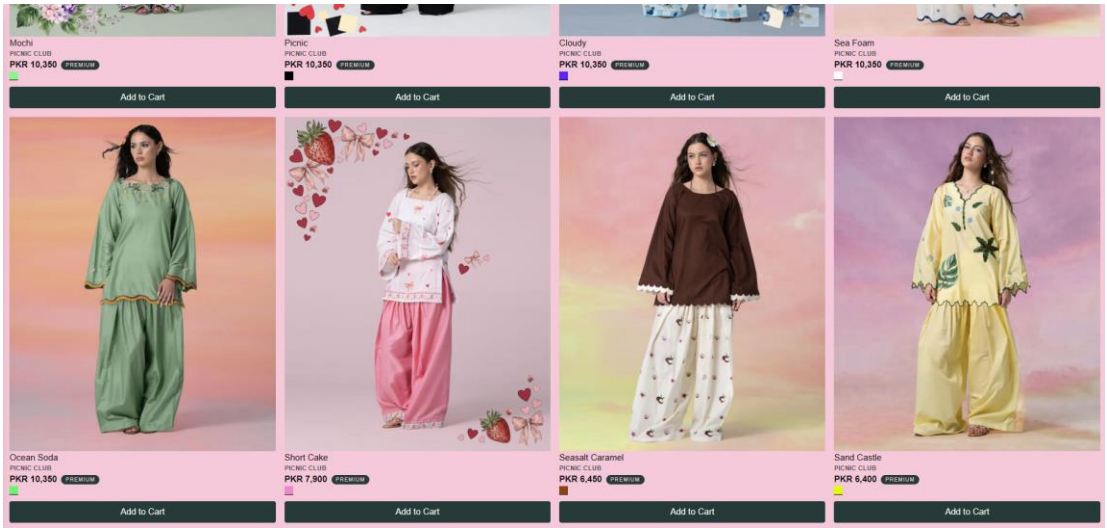
[screenshot of Task 3 at 1280px]



[without sort]



[1st sort: low to high]



[2nd sort: high to low]

Code added or changed

HTML - search box, sort button and "No products found" message in picnicclub.html:

```
<!-- Task 3: search + sort controls -->
      <div class="flex flex-wrap items-center gap-3 mb-4">
        <label for="search-box" class="sr-only">Search products</label>
        <input
          type="search"
          id="search-box"
          placeholder="Search products..."
          class="border border-gray-400 rounded px-4 py-2 bg-white w-full sm:w-72"
        />
        <button
          type="button"
          id="sort-btn"
          class="bg-[#263b39] text-white py-2 px-5 rounded hover:bg-[#405b57]"
        >
          Sort by price
        </button>
      </div>
```

```
      <p id="no-results" class="is-hidden text-center text-lg font-semibold text-[#263b39] py-
10">
        No products found
      </p>

      <section id="product-grid"
```

CSS helper classes in picnicclub.html:

```
.search-hidden,
      .is-hidden {
        display: none !important;
      }
      .premium-label {
        display: inline-block;
        margin-left: 0.5rem;
        padding: 1px 8px;
        border-radius: 9999px;
        background: #263b39;
        color: #f5c9da;
        font-size: 10px;
        font-weight: 400;
        letter-spacing: 1px;
        text-transform: uppercase;
        vertical-align: middle;
      }
```

script.js:

```
// Midterm - Task 3: live search, price sort, Premium label
// N = 15  ->  Premium limit = N x 100 = 1500

const N = 15;
const PREMIUM_LIMIT = N * 100;

const grid = document.getElementById("product-grid");
const cards = Array.from(grid.querySelectorAll("article"));
const searchBox = document.getElementById("search-box");
const sortBtn = document.getElementById("sort-btn");
const noResults = document.getElementById("no-results");
const categoryRadios = document.querySelectorAll('input[name="category"]');

let ascending = true; // first click = lowest to highest

// read the price from the card itself ("PKR 10,350" -> 10350)
function getPrice(card) {
  const text = card.querySelector(".product-price").textContent;
  return parseInt(text.replace(/[^0-9]/g, ""), 10);
}

// a card counts as visible if neither the category filter (CSS)
// nor the search (class) has hidden it
function updateNoResults() {
  const anyVisible = cards.some(
    (card) => getComputedStyle(card).display !== "none",
  );
  noResults.classList.toggle("is-hidden", anyVisible);
  grid.classList.toggle("is-hidden", !anyVisible);
}

// search by product name, not case-sensitive
function applySearch() {
  const term = searchBox.value.trim().toLowerCase();
  cards.forEach((card) => {
    const name = card.querySelector(".product-name").textContent.toLowerCase();
    card.classList.toggle("search-hidden", !name.includes(term));
  });
  updateNoResults();
}

searchBox.addEventListener("input", applySearch);

// category radios are pure CSS, but the message must refresh when they change
```

```
categoryRadios.forEach((radio) => {
  radio.addEventListener("change", updateNoResults);
});

// sort by price: move the existing card elements with appendChild
sortBtn.addEventListener("click", () => {
  const sorted = [...cards].sort((a, b) =>
    ascending ? getPrice(a) - getPrice(b) : getPrice(b) - getPrice(a),
  );
  sorted.forEach((card) => grid.appendChild(card));


  sortBtn.textContent = ascending
    ? "Sort by price: low to high"
    : "Sort by price: high to low";
  ascending = !ascending;
});

// Premium label, added once when the page loads
cards.forEach((card) => {
  if (getPrice(card) > PREMIUM_LIMIT) {
    const label = document.createElement("span");
    label.className = "premium-label";
    label.textContent = "Premium";
    card.querySelector(".product-price").appendChild(label);
  }
});
```